

Stochastic Methods in Finance

Assignment

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1 Introduction

2 Approximation

The evolution of the stock price can also be expressed in terms of log-returns. Supposed we observe the price of the stock at discrete time points over the interval $[0, T]$, where observations are taken at regular intervals of size Δ_t . Specifically, we observe prices $Y_{t_i} = \log(S_{t_i})$ at times $t_i = i\Delta_t$ for $i = 0, 1, \dots, n$, where $n\Delta_t = T$. Then $r_{t_i} = Y_{t_i} - Y_{t_{i-1}}$ is the continuously compounded return of the stock over the period $[t_{i-1}, t_i]$. The evolution of the log price can therefore be written as

$$Y_{t_j} = Y_{t_0} + \sum_{i=1}^j r_{t_i} \quad j = 1, \dots, n. \quad (1)$$

The returns r_{t_i} have the following moments

$$\begin{aligned} E(r_{t_i}) &= \frac{1}{2}\sigma\sqrt{\Delta_t} - \frac{1}{2}\sigma\sqrt{\Delta_t} = 0 \\ \text{Var}(r_{t_i}) &= \frac{1}{2}\sigma^2\Delta_t + \frac{1}{2}\sigma^2\Delta_t = \sigma^2\Delta_t. \end{aligned} \quad (2)$$

where σ is the annualized volatility, which is assumed to constant. Since the returns are assumed to be independent, we can approximate the distribution of the return over an entire period with a normal distribution thanks to the central limit theorem

$$Y_{t_j} - Y_{t_0} = \sum_{i=1}^j r_{t_i} \sim N(0, j\sigma^2\Delta_t). \quad (3)$$

From that it follows that the price S_{t_j} can be approximated as

$$\begin{aligned} S_{t_j} &\approx \exp(Y_{t_0} + N(0, j\sigma^2\Delta_t)) \\ &= S_{t_0} \cdot \exp(N(0, j\sigma^2\Delta_t)) \end{aligned} \quad (4)$$

and hence the average price over the entire period can be approximated as

$$\begin{aligned} \bar{S} &\approx \frac{1}{n} \sum_{j=1}^n S_{t_0} \exp(N(0, j\sigma^2\Delta_t)) \\ &= \frac{S_{t_0}}{n} \sum_{j=1}^n \exp(N(0, j\sigma^2\Delta_t)) \end{aligned} \quad (5)$$